Project Specifications

PROPCHAIN-AI

2 User and System Specifications Table

Table 1: User and System Specifications Table

Parameters	Specifications
System Type	AI-assisted property document analysis, ownership-chain verification, risk assessment, and blockchain integrity platform.
Target Users	Property buyers, owners, legal/review professionals, property-service users, and administrators.
Primary Interaction	Web-based document upload, structured property details, dashboards, ownership graph, verification results, and report generation.
Recognition Method	OCR, document classification, NLP entity extraction, normalization, cross-document comparison, and rule-based analysis.
User Dependency	User supplies property documents and property details; the system analyzes only the evidence and data sources available to it.
Maximum Users Supported	Not fixed for the academic prototype; designed as a modular web application that can be scaled through backend and database deployment.
Supported Interaction Techniques	File upload, form input, dashboard interaction, graph/timeline visualization, search, review actions, and AI-assisted question answering.
Document Requirement	PDF/image property records, preferably clear scans or digitally generated documents; multilingual support is planned initially for English and Marathi.
Processing Type	Batch document analysis with near-real-time structured comparison and risk reporting after processing.
Application Areas	Property due diligence, document consistency checking, ownership-history analysis, land-record analysis, RERA screening, and document-integrity verification.
Security Compliance	Authentication, authorization, encrypted storage, access control, audit logging, privacy-aware design, and avoidance of storing sensitive document contents on blockchain.

3 Input Specifications Table

Table 2: Input Specifications Table

Parameters	Specifications
Input Type	Property documents, scanned PDFs/images, structured property information, and optional verification/source information.
Input Source	User-uploaded sale deeds, agreements, 7/12, 8A, Ferfar, Property Card, EC and other supporting documents.
Supported Languages	English and Marathi for the initial prototype; Hindi and additional Indian languages can be added later.
File Formats	PDF, JPG, JPEG, and PNG for the prototype; exact upload limits are configurable.
Image Quality	Clear, readable scans suitable for OCR; preprocessing may be applied for skew, noise, contrast, and orientation.
Text Input	Structured values such as property ID, state, district, taluka, village, survey/Gat/CTS number, area, and transaction details.
Property Data	Owner/seller/buyer names, dates, property identifiers, area, boundaries, transaction type, registration information, and other extracted entities.
Language Detection	Automatic or configurable detection before OCR/NLP processing.
OCR Method	OCR pipeline for scanned documents, followed by text cleaning and structured entity extraction.
Document Classification	Classification into categories such as Sale Deed, Agreement to Sale, Gift Deed, Mortgage, Lease, Partition, Will, 7/12, 8A, Ferfar, Property Card, EC, RERA, court/supporting document, and other relevant types.

4 AI Document Analysis Specifications Table

Table 3: AI Document Analysis Specifications Table

Parameters	Specifications
Document Classification	Identify the likely document type from uploaded PDF/image content before downstream analysis.
OCR Processing	Convert scanned property documents into searchable text and preserve page-level evidence where possible.
Entity Extraction	Extract names, dates, survey/Gat/CTS numbers, ULPIN where available, area, boundaries, registration numbers, consideration/value references, parties, and document references.
Name Normalization	Normalize punctuation, abbreviations, transliteration and spelling variations while treating similarity as an indication rather than proof of identity.
Property Identifier Normalization	Map equivalent forms such as Gat No., Gat Number, Survey No., CTS No. and other jurisdiction-specific identifiers into canonical fields.
Document Comparison	Compare extracted values across documents for owner names, property identifiers, area, boundaries, dates, transaction relationships, and registration information.
Chronology Analysis	Check temporal consistency among execution, registration, death, mutation, mortgage, release, transfer and other documented events.
Missing-Link Detection	Identify references to predecessor deeds, mutation records, supporting documents, approvals, or other records that were referenced but not supplied.
Anomaly Detection	Flag unusual or conflicting patterns such as duplicate transfers, ownership-chain gaps, area discrepancies, conflicting encumbrance statements, or suspicious chronology.
Explainable AI	Each finding should provide the reason, evidence/document page where available, confidence indicator, and suggested verification action.
Legal Rule Matching	Evaluate extracted facts against configurable Central and state-specific rules without presenting the result as a conclusive legal determination.
RAG / AI Assistant	Answer user questions from approved legal sources and analyzed property documents, with source-aware responses and uncertainty handling.
Human Review	Allow authorized reviewers to inspect, comment on, resolve, or mark AI findings for further verification without deleting the original audit record.

5 Property Verification Workflow Specifications Table

Table 4: Property Verification Workflow Specifications Table

Parameters	Specifications
Request Type	Create property, upload documents, analyze documents, compare records, view ownership history, verify integrity, review findings, and generate report.
Input Format	PDF/image documents plus structured property information entered through secure web forms.
Required Information	Minimum property identity plus available supporting documents; exact document set depends on property type and jurisdiction.
Document Processing	Upload → validation → OCR/classification → entity extraction → normalization → storage of structured results.
Ownership Analysis	Construct a chronological ownership/transaction graph such as A → B → C → D from supplied evidence.
Cross-Document Check	Compare owners, seller/buyer relationships, survey/Gat/CTS identifiers, area, boundaries, dates, registrations, and references.
Legal Screening	Apply the configured jurisdictional rule pack covering relevant transfer, registration, land-record, succession, RERA, encumbrance, planning, and other applicable checks.
Risk Assessment	Generate category-wise findings and an application-defined risk indicator using severity, evidence and confidence.
Blockchain Operation	Generate SHA-256 fingerprint for each finalized digital document and record the

Parameters	Specifications
	fingerprint, document ID, timestamp, and verification event on the blockchain.
Integrity Verification	Recalculate the uploaded file hash and compare it with the recorded blockchain fingerprint to return MATCH or MISMATCH.
Conflict Handling	Flag contradictions for review rather than automatically declaring a document fraudulent or title invalid.
Processing Method	Next.js → FastAPI → OCR/NLP + Rule Engine → PostgreSQL/Object Storage → Blockchain → FastAPI → Dashboard/Report.
Record Storage	Structured property/analysis data in PostgreSQL; encrypted document storage outside blockchain; audit and verification metadata retained separately.

6 Model and Technology Specifications Table

Table 5: Model and Technology Specifications Table

Parameters	Specifications
Document AI	OCR + document classification + layout/text processing for property-paper understanding.
OCR / Speech-to-Text	Not applicable as a core component; the system is document-focused and uses OCR for scanned property records.
NLP / Entity Understanding	spaCy/Transformers-based NLP or equivalent models for named-entity extraction, relation extraction, normalization and document understanding.
Document Similarity	Sentence-transformer embeddings or comparable semantic similarity method for comparing related documents and references.
Anomaly Detection	Rule-based consistency checks with optional ML anomaly detection for unusual property-document patterns.
Legal Knowledge Layer	Versioned rule database containing jurisdiction, law/section, conditions, severity, evidence requirement, effective dates, and source reference.
Ownership Graph	Graph representation of persons/entities, properties, documents and transfer relationships; React Flow visualization for the prototype.
Risk Engine	Deterministic rule-based scoring using finding type, severity, confidence and evidence; displayed as a project-defined indicator.
Backend Framework	FastAPI (Python).
Database Engine	PostgreSQL for structured records; object storage or secure local storage for encrypted document files.
Vector / Retrieval Layer	FAISS or an equivalent vector-retrieval component may be used for semantic legal/document search in the AI assistant.
Blockchain	Solidity-based smart contract on a local/private EVM-compatible network for document hashes and verification events.
API Style	RESTful APIs between frontend, backend, AI services, storage and blockchain services.
Frontend Framework	Next.js with TypeScript, Tailwind CSS, dashboards, document comparison views, timeline and ownership graph.
Training Strategy	Initial prototype may use pre-trained OCR/NLP models plus a small labeled/synthetic dataset; custom fine-tuning is optional and depends on dataset quality.
Deployment	Development on Windows/Linux/macOS; production deployment can use containerized backend, database, storage and blockchain services.

7 Output Specifications Table

Table 6: Output Specifications Table

Parameters	Specifications
Output Type	AI-assisted property due-diligence result, ownership-chain visualization, document-integrity status, risk findings, and final report.
Document Analysis Output	Document type, extracted entities, page-level evidence, confidence indicators, and structured property information.
Ownership Output	Chronological ownership/transaction graph and timeline derived from the supplied document set.
Consistency Output	Match/mismatch/unknown results for owner names, property identifiers, area, dates, boundaries, registration details and other configured fields.
Risk Output	Category-wise findings covering ownership, identity, area, encumbrance, mortgage, succession, litigation, land-use, RERA, planning, document-integrity and missing-document indicators.
Blockchain Output	Document ID, SHA-256 fingerprint, registration timestamp, transaction reference, and MATCH/MISMATCH verification result.
Legal Screening Output	Applicable rule references, triggered conditions, source/evidence references, and suggested verification actions.
Review Output	Reviewer comments, resolution status, additional-document requests, and immutable audit history of AI findings.
Report Output	Property Due-Diligence Report containing property identity, documents analyzed, ownership chain, findings, risk summary, blockchain verification, sources and limitations.
AI Assistant Output	Natural-language answers grounded in configured legal sources and the analyzed document set, with explicit uncertainty where evidence is incomplete.
Processing Mode	Document processing in batch/near-real-time depending on file size and AI/OCR workload.
Export Formats	PDF report and downloadable structured analysis data; additional formats may be added later.
Integration Support	Next.js ↔ FastAPI REST APIs ↔ PostgreSQL/Object Storage ↔ AI/NLP services ↔ Blockchain smart contract.
Important Limitation	The system provides documentary analysis and risk indicators. It does not establish legal title, certify ownership, replace government records, or constitute legal advice.

Project Team: 1. Nadaf MohammadAzhar

2. MohammadAli Shaikh

3. Vishal Sahani